

Below are **10 Real Industry Scenario-Based Linux Tasks** with:

- ✓ **Script provided**
- ✓ **Required package installation**
- ✓ **How to run manually**
- ✓ **Student task: Create the cron job**

Students only need to **write the correct crontab entry**.

1 Daily MySQL Database Backup

Install Required Package

```
dnf install mysql -y
```

Script

Create file:

```
nano /scripts/db_backup.sh
```

Script:

```
#!/bin/bash
```

```
DATE=$(date +%F)
```

```
BACKUP_DIR="/backup"
```

```
mkdir -p $BACKUP_DIR
```

```
mysqldump -u root -pPassword mydb > $BACKUP_DIR/mydb_$(date +%F).sql
```

Make executable:

```
chmod +x /scripts/db_backup.sh
```

Test Script

/scripts/db_backup.sh

Student Task

Create a cron job to run this script daily at 2 AM.

2 Delete Old Log Files

Script

```
nano /scripts/log_cleanup.sh
#!/bin/bash
```

```
LOG_DIR="/var/log/myapp"
```

```
find $LOG_DIR -type f -mtime +10 -delete
```

Make executable:

```
chmod +x /scripts/log_cleanup.sh
```

Test Script

/scripts/log_cleanup.sh

Student Task

Create a cron job to run this script daily at 3 AM.

3 Disk Usage Monitoring

Script

```
nano /scripts/disk_monitor.sh
```

```
#!/bin/bash
```

```
DATE=$(date)
```

```
echo "Disk usage check at $DATE" >> /var/log/disk_usage.log
```

```
df -h >> /var/log/disk_usage.log
```

```
echo "-----" >> /var/log/disk_usage.log
```

Make executable:

```
chmod +x /scripts/disk_monitor.sh
```

Test Script

```
/scripts/disk_monitor.sh
```

Student Task

Create a cron job to run this script every 15 minutes.

4 Restart Apache Web Server

Install Apache

```
dnf install httpd -y
```

Script

```
nano /scripts/restart_apache.sh
```

```
#!/bin/bash
```

```
systemctl restart httpd
```

Make executable:

```
chmod +x /scripts/restart_apache.sh
```

Test Script

```
/scripts/restart_apache.sh
```

Student Task

Create a cron job to restart Apache daily at 4 AM.

5 Website Health Check

Install Curl

```
dnf install curl -y
```

Script

```
nano /scripts/site_check.sh
#!/bin/bash
```

```
SITE="https://google.com"
```

```
curl -s $SITE > /dev/null
```

```
if [ $? -eq 0 ]
then
    echo "$(date) : Website is UP" >> /var/log/site_status.log
else
    echo "$(date) : Website is DOWN" >> /var/log/site_status.log
fi
```

Make executable:

```
chmod +x /scripts/site_check.sh
```

Test Script

```
/scripts/site_check.sh
```

Student Task

Create a cron job to check website every 5 minutes.

6 Backup Files to Remote Server

Install rsync

```
dnf install rsync -y
```

Script

```
nano /scripts/file_backup.sh  
#!/bin/bash
```

```
rsync -av /data backup@192.168.1.20:/backup/
```

Make executable:

```
chmod +x /scripts/file_backup.sh
```

Test Script

```
/scripts/file_backup.sh
```

Student Task

Create a **cron job** to run the backup daily at 2:30 AM.

7 Temporary File Cleanup

Script

```
nano /scripts/tmp_cleanup.sh  
#!/bin/bash
```

```
find /tmp -type f -mtime +5 -delete
```

Make executable:

```
chmod +x /scripts/tmp_cleanup.sh
```

Test Script

/scripts/tmp_cleanup.sh

Student Task

Create a cron job to run daily at 1 AM.

8 Security Updates Automation

Script

```
nano /scripts/system_update.sh
#!/bin/bash
```

```
dnf update -y
```

Make executable:

```
chmod +x /scripts/system_update.sh
```

Test Script

```
/scripts/system_update.sh
```

Student Task

Create a cron job to run every Sunday at 3 AM.

9 Generate Daily System Report

Script

```
nano /scripts/system_report.sh
#!/bin/bash
```

```
DATE=$(date +%F)
```

```
echo "System Report $DATE" > /var/log/system_report_$DATE.log
```

```
uptime >> /var/log/system_report_$DATE.log
```

```
df -h >> /var/log/system_report_$DATE.log
```

```
free -h >> /var/log/system_report_$DATE.log
```

Make executable:

```
chmod +x /scripts/system_report.sh
```

Test Script

```
/scripts/system_report.sh
```

Student Task

Create a **cron job** to **generate report daily at midnight**.

Run Script Once Using **at**

Script

```
nano /scripts/restart_service.sh
```

```
#!/bin/bash
```

```
systemctl restart httpd
```

```
echo "Service restarted at $(date)" >> /var/log/service_restart.log
```

Make executable:

```
chmod +x /scripts/restart_service.sh
```

Test Script

```
/scripts/restart_service.sh
```

Student Task

Schedule this script to **run once after 20 minutes** using the **at** command.

These Tasks Teach Real Industry Skills

Students learn:

- ✓ Automation
- ✓ Monitoring
- ✓ System maintenance
- ✓ Backup strategies
- ✓ Service management
- ✓ Scheduling jobs

These are **core skills of Linux Admins and DevOps Engineers**.
